

Predicting NYC High School Performance

Advanced Data Analytics
Vi Kovacevic

Project Overview

- Objective: Predict NYC high school Regents Test performance (Low / Medium / High)
- Need: Overall Decline in NYS Student Performance
- Original Dataset: NYC Public High Schools Regents Scores
- Inspiration: [Catholic Schools and the Common Good, Anthony S. Bryk](#)
- Methods: Supervised learning (KNN, Random Forest) + Clustering + PCA
- Hypothesis: Teacher to Pupil ratios will have the strongest prediction power for student performance

Loaded 410 rows and 20 columns

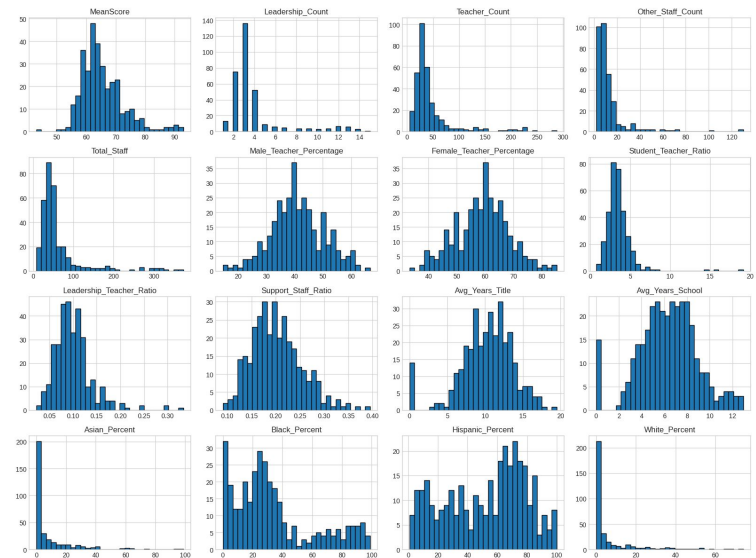
	DBN	School_Name	Neighborhood	MeanScore	Leadership_Count	Teacher_Count	Other_Staff_Count	Total_Staff	Male_Teacher_Percentage	Female_Teacher_Percentage	Student_Teacher_Ratio	Leadership_Teacher_Ratio
0	02M260	THE CLINTON SCHOOL	Pelham - Throgs Neck	66.843889	3.0	56.0	11.0	70.0	23.214286	76.785714	1.292659	0.053571
1	02M413	SCHOOL OF THE FUTURE HIGH SCHOOL	Bensonhurst - Bay Ridge	70.226190	4.0	51.0	17.0	72.0	43.137255	56.862745	14.581699	0.078431
2	02M422	QUEST TO LEARN	Chelsea - Clinton	66.642111	3.0	51.0	16.0	70.0	35.294118	64.705882	1.743945	0.058824
3	02M546	ACADEMY FOR SOFTWARE ENGINEERING	Union Square - Lower East Side	69.660333	3.0	32.0	13.0	48.0	46.875000	53.125000	3.556641	0.093750
4	03M860	FREDERICK DOUGLASS ACADEMY II SECONDARY SCHOOL	Bedford Stuyvesant - Crown Heights	62.231222	3.0	38.0	11.0	52.0	39.473684	60.526316	1.329721	0.078947

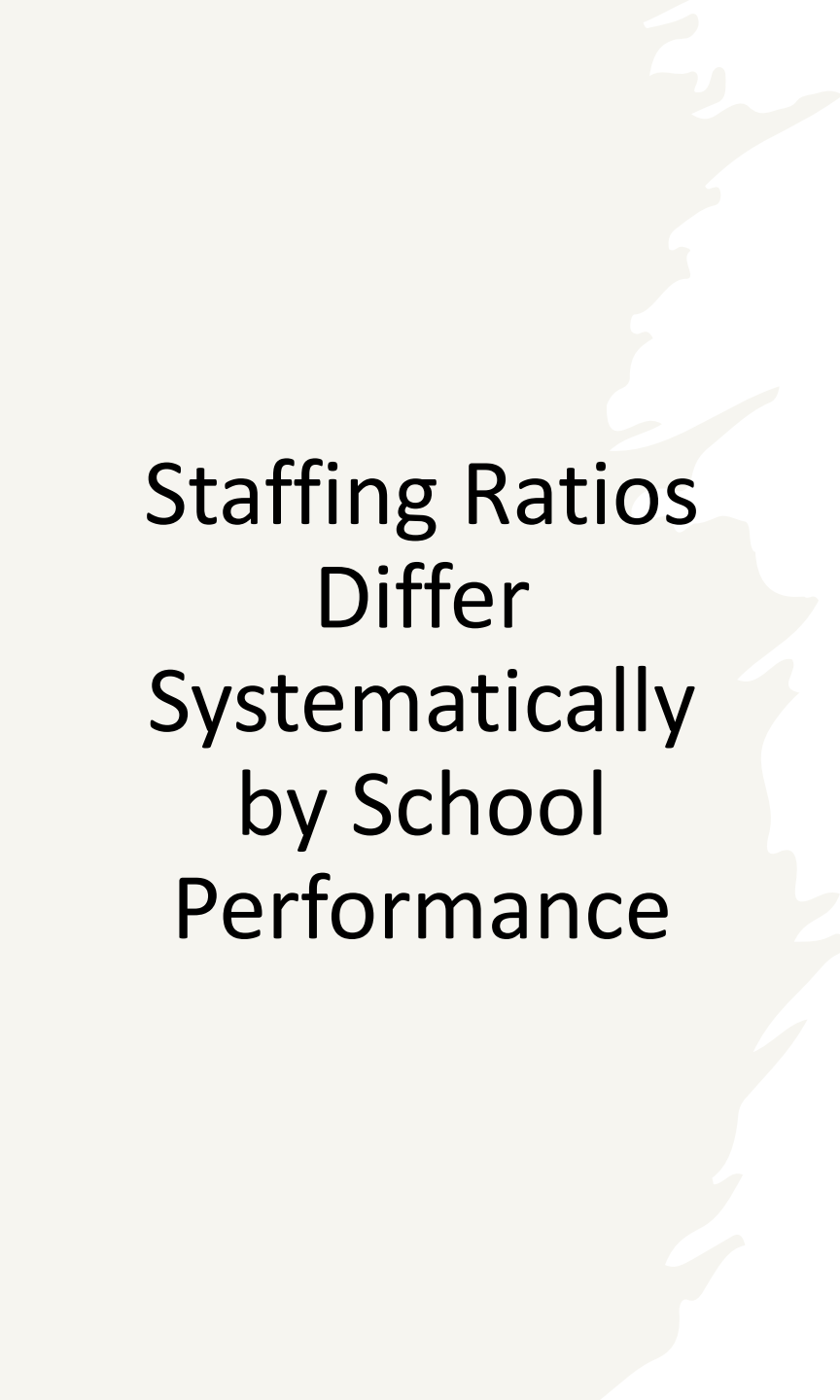
Data Overview

- **410 NYC public high schools**
- **Mean student–teacher ratio: 3.47** (median **3.23**), with wide variation
- **Ratios range from 0.74 to 19.16**, indicating substantial inequality in instructional capacity
- **Leadership and support staff ratios also vary**, reflecting differences in institutional structure
- **Cons from Data Compression:** Lost the ability to see how a *specific* test type changed in a *specific* school over time.

Feature Distributions & Modeling Implications

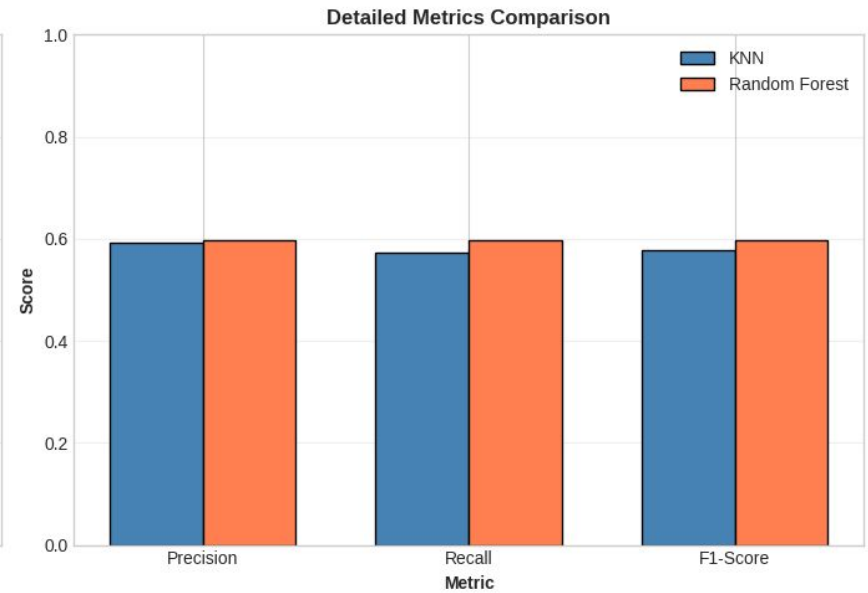
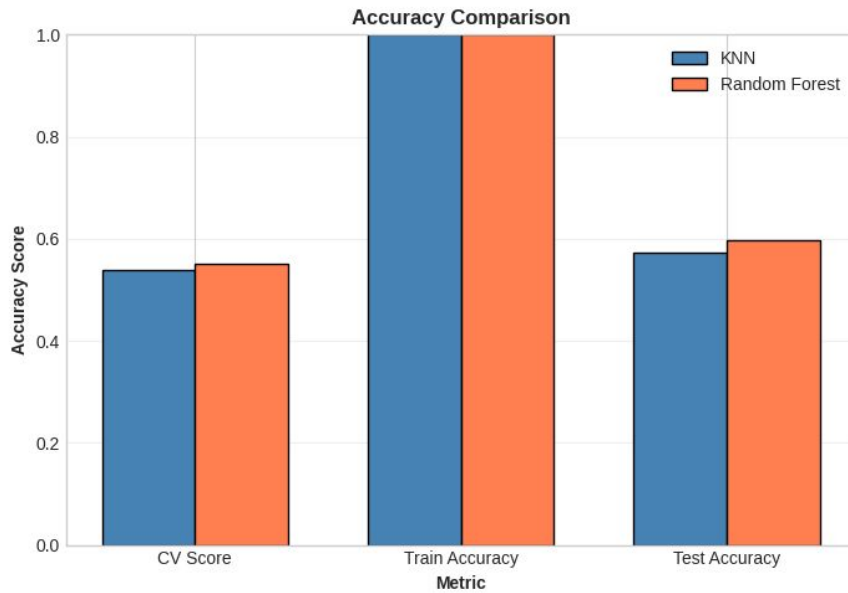
- Many staffing and demographic features are **highly skewed**, reflecting unequal school scale
- **Staffing ratios** are **more stable** than raw counts, making them more informative predictors
- These patterns explain **moderate prediction accuracy** and **weak clustering performance**





Staffing Ratios Differ Systematically by School Performance

- **High-performing schools:** mean student–teacher ratio **3.07**
- **Medium-performing schools:** **3.44**
- **Low-performing schools:** **3.89**
- High-performing schools employ **more total staff**, but maintain **more favorable ratios**
- Differences suggest **resource allocation**, not raw staff counts, drives performance patterns

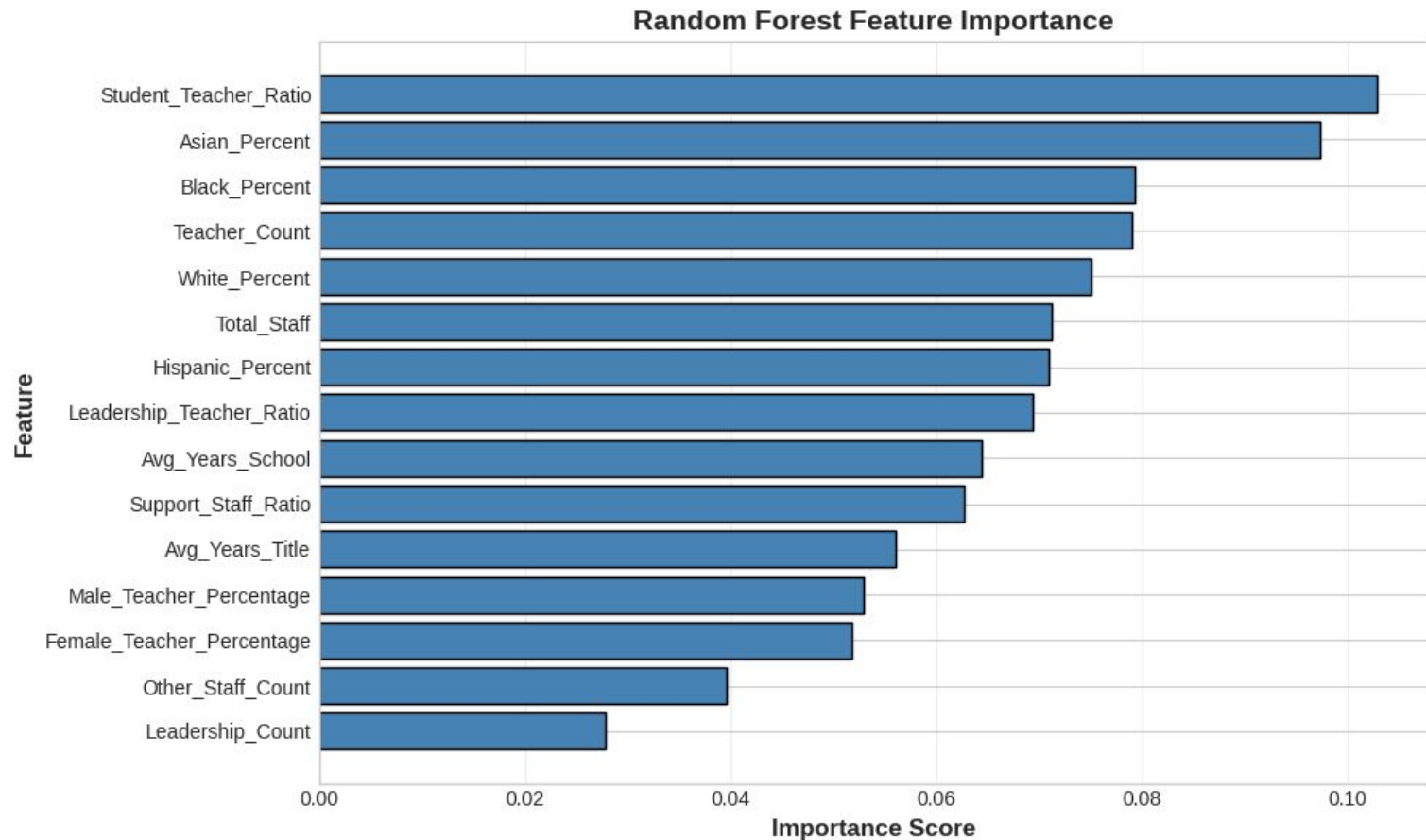


Model Accuracy Comparison

- Random Forest consistently outperforms KNN on test data.
- KNN shows high variance and overfitting at low k values.

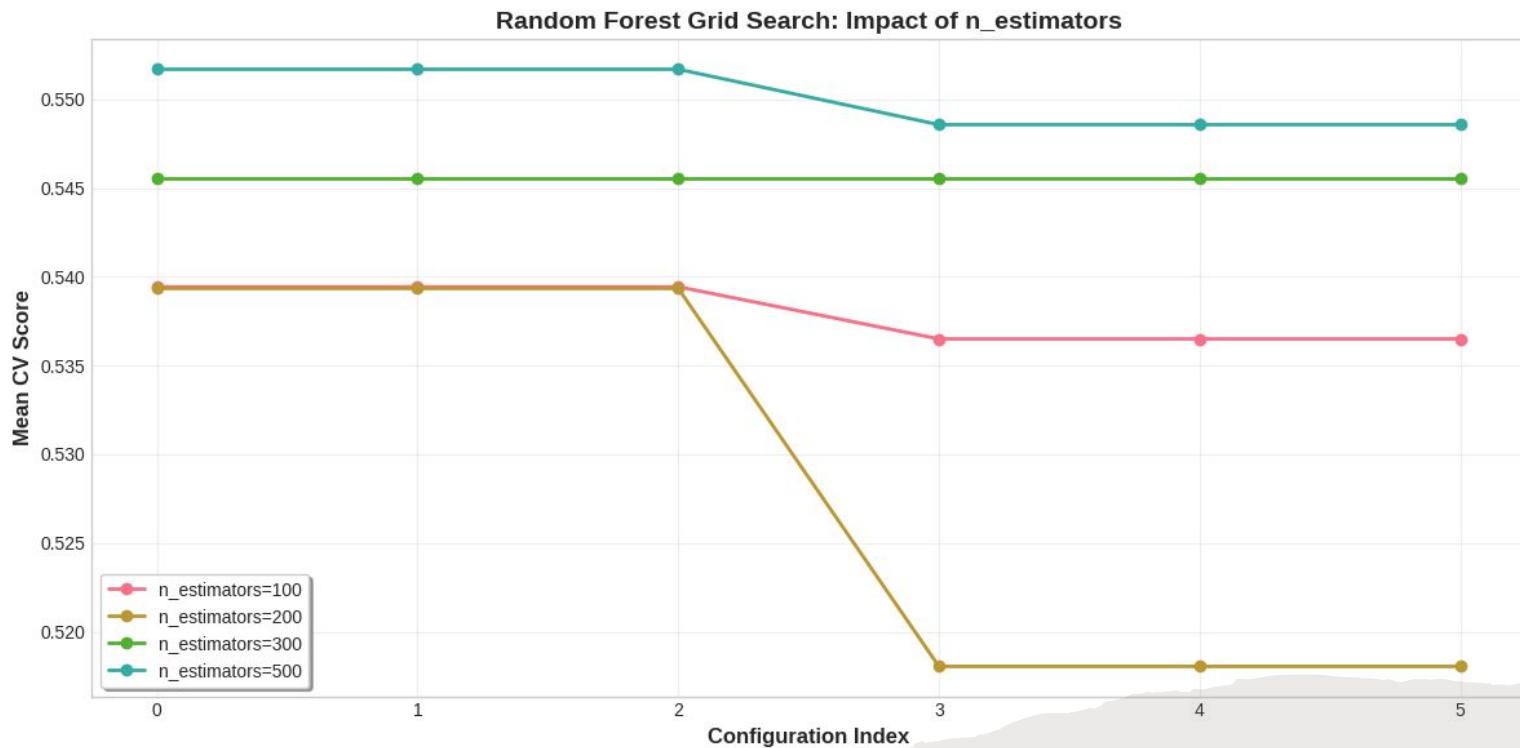
Random Forest Feature Importance

Staffing ratios and student demographics are the strongest predictors.



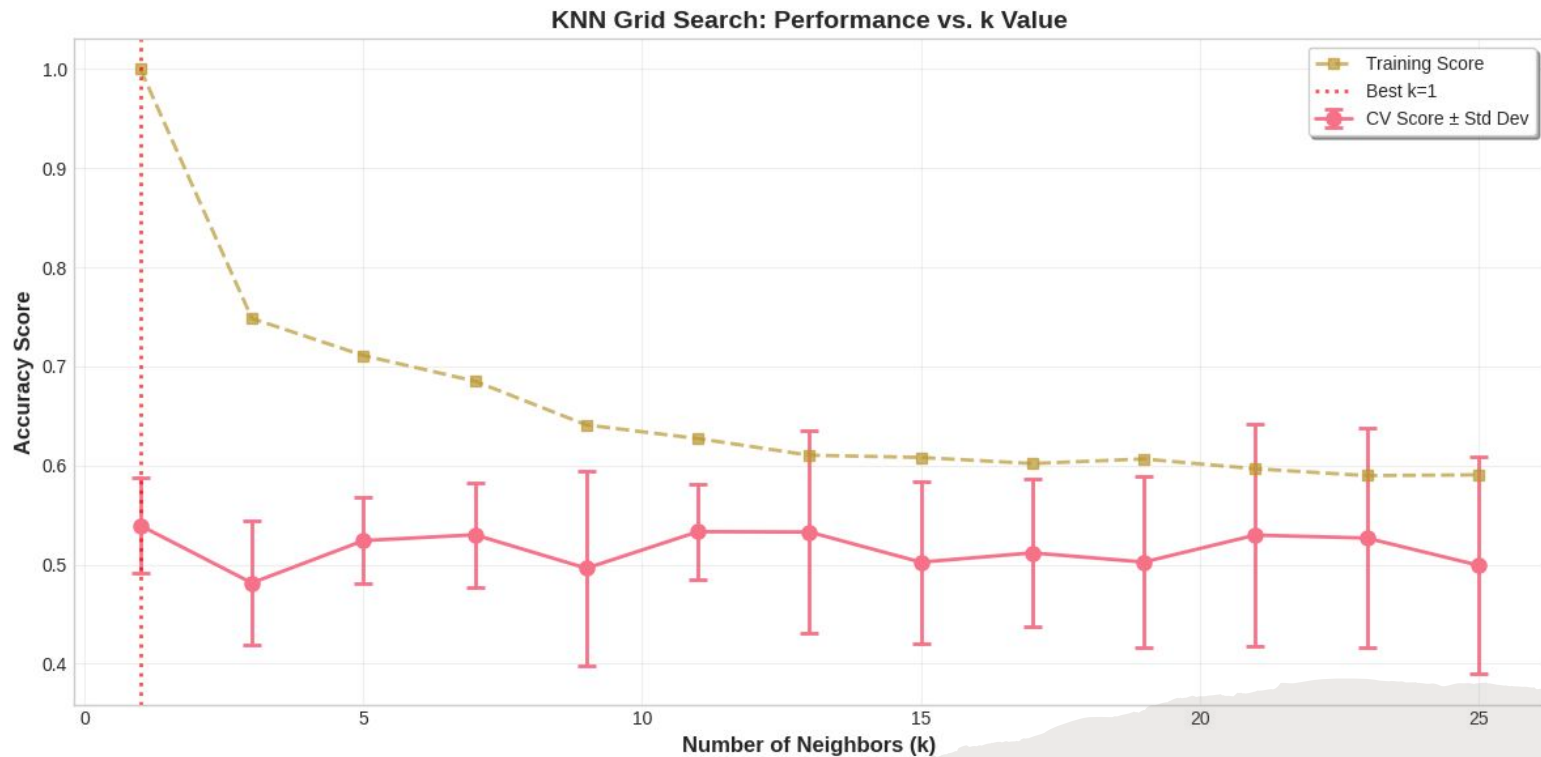
Random Forest Grid Search

- Increasing `n_estimators` improves stability.
- Best performance achieved around 500 trees.

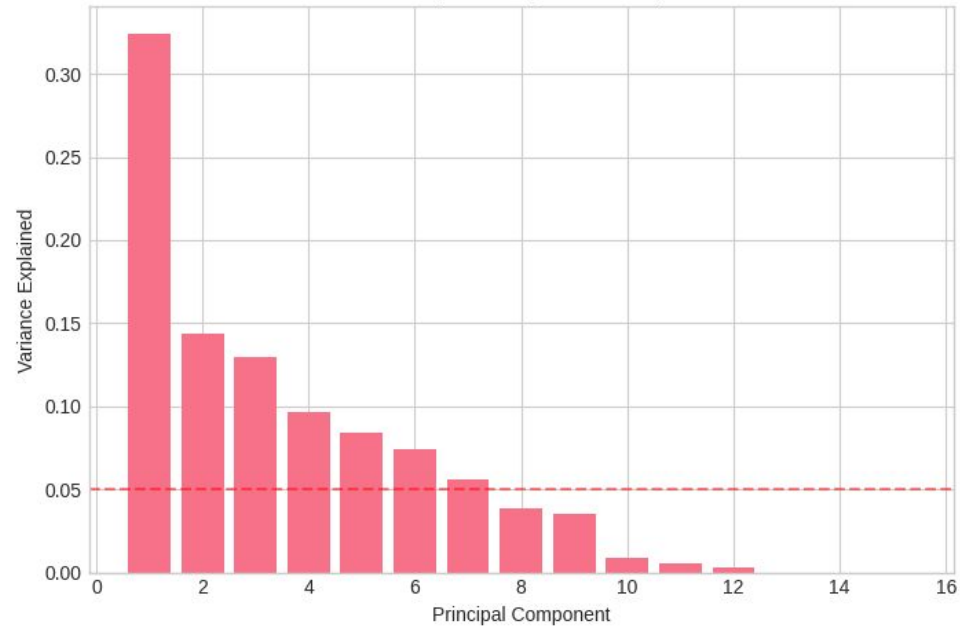


KNN Hyperparameter Tuning

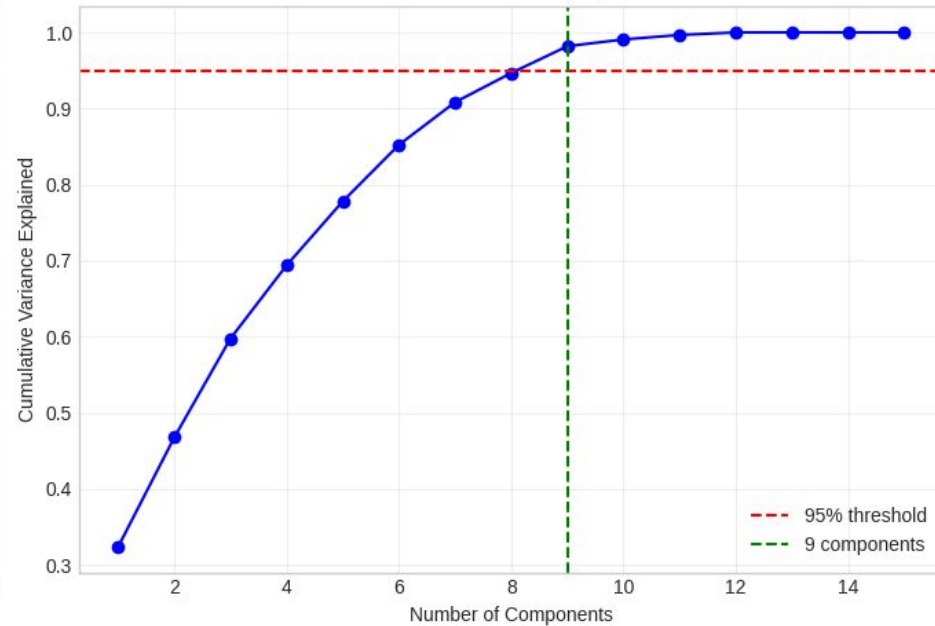
- $k=1$ maximizes CV score but severely overfits.
- Higher k values trade accuracy for stability.



Variance Explained by Each Component

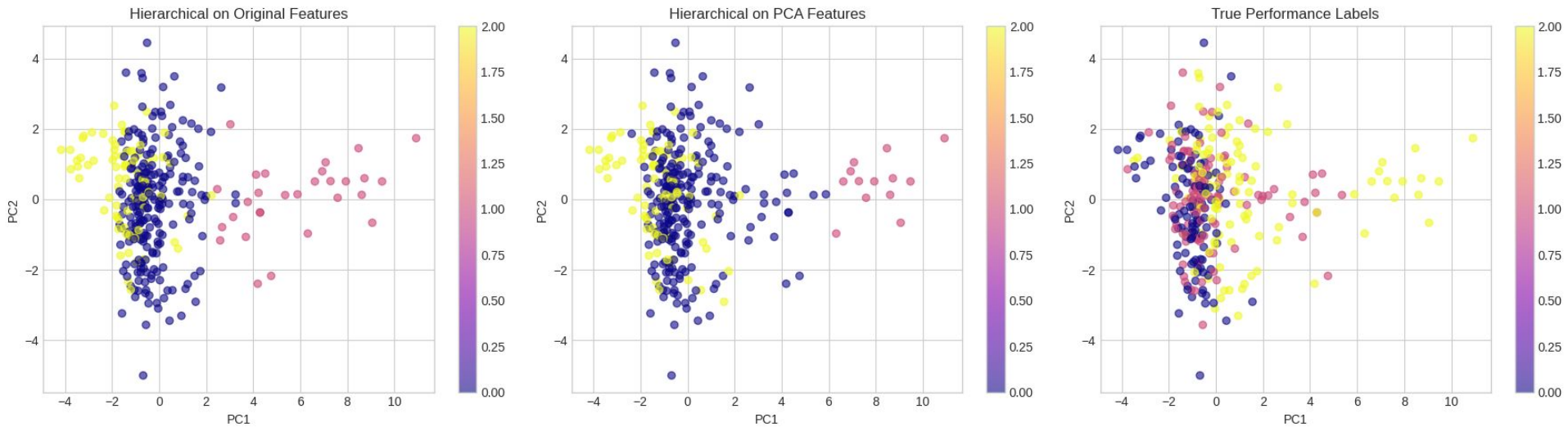


Cumulative Variance Explained



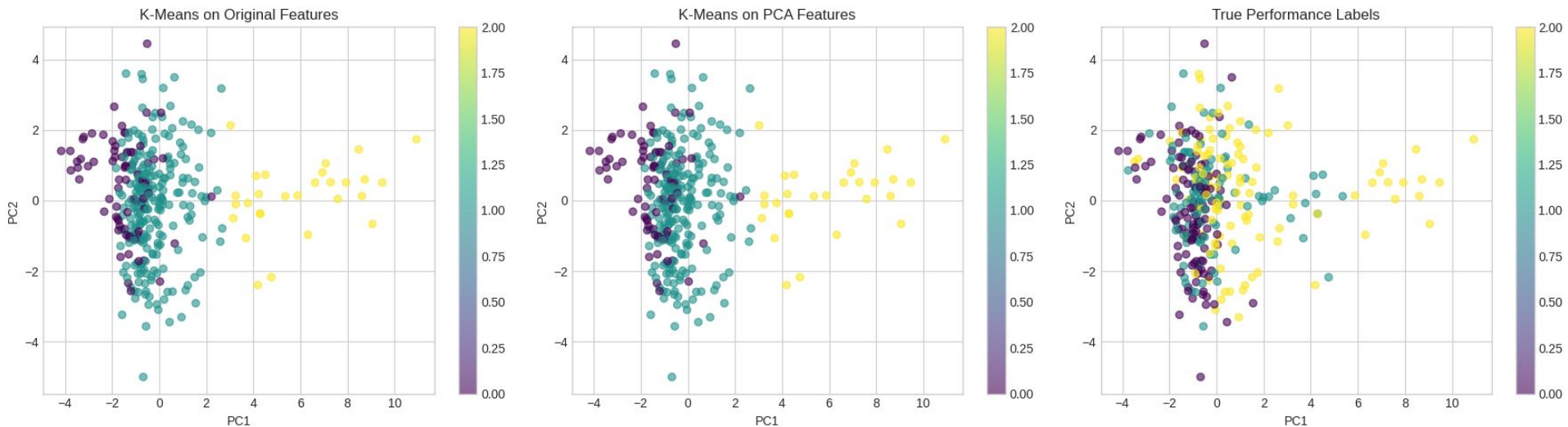
Dimensionality Reduction (PCA)

- 9 components explain ~95% of total variance.
- PCA used to improve clustering visualization.



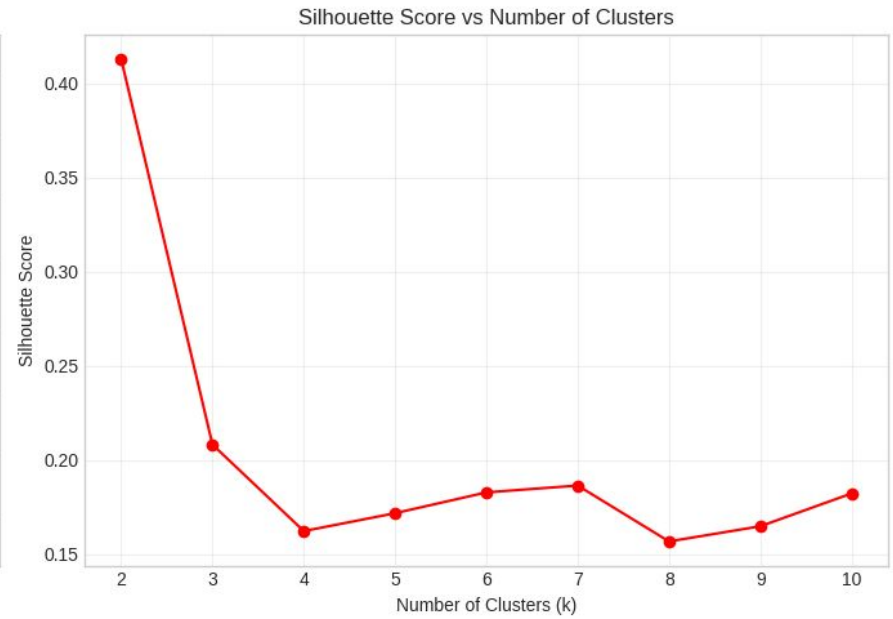
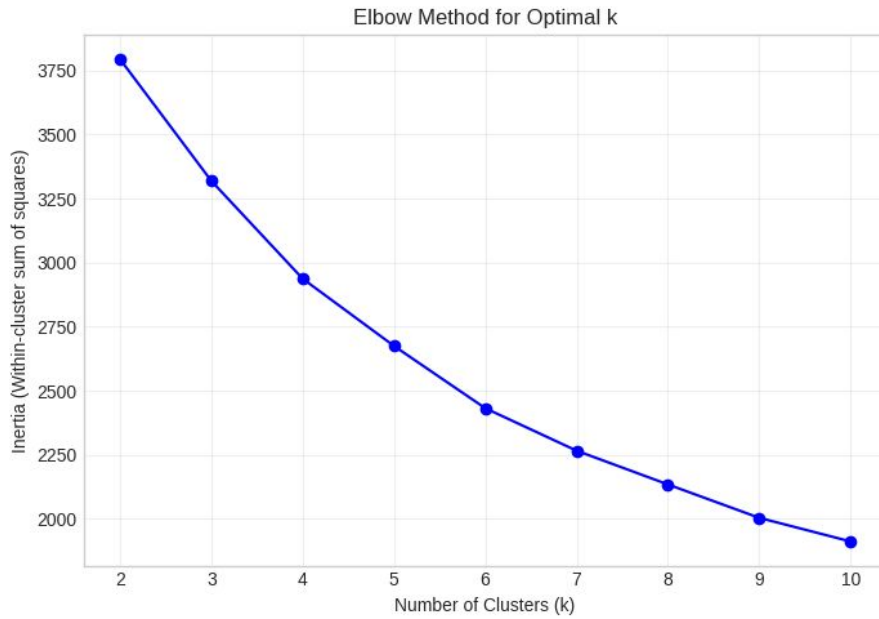
Hierarchical Clustering Results

- Clusters show partial alignment with performance labels.
- Some overlap indicates structural similarity across groups.



K-Means Clustering Results

- Best clustering occurs at $k \approx 3-4$.
- Clusters partially correspond to Low / Medium / High categories.



Optimal Cluster Selection

- Elbow and silhouette methods suggest $k=3$ as optimal.
- Matches known performance categories.

Clustering Results and Limitations

- Unsupervised clustering produced **weak separation** across performance categories
- **Substantial overlap** exists between clusters, even after PCA
- School performance does not fall into clearly distinct groups
- Clustering difficulty reflects system-wide inequality

Key Takeaways

- Random Forest is the strongest classifier
- Staffing ratios matter more than raw counts
- Performance categories are partially separable
- Structural inequality is reflected in the data

Expanded Takeaways

- School performance is strongly shaped by structural conditions
- **Staffing ratios act as a proxy for resource access**, administrative capacity, and institutional support
- **Raw staff counts are less informative than how resources are distributed**, highlighting inefficiencies and systemic imbalance

Expanded Expanded Takeaways

- The results **support the hypothesis** that staffing ratios have the strongest predictive power for student performance.
- Staffing ratios outperform raw staffing counts, indicating that **how resources are allocated matters more than quantity alone**.
- Remaining overlap in performance categories reflects **structural inequality across NYC public schools**, shaped by policy and neighborhood context.

Thank you!

[GitHub](#)